Emotion Classification Pipeline

Samara · Ahmed · Barsam
BUas – Comp Sci & AI

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Overview



Objective: Build and test an emotion-analysis pipeline to uncover challenges and insights that can help our client enhance their existing system.



Chosen show

Why Australian Survivor?

- Emotion-rich content.
- Diverse interactions.

Transcript Extraction

Accuracy Comparison (WER)

- **AssemblyAI:** 3.84%
- **Whisper:** 5.02%

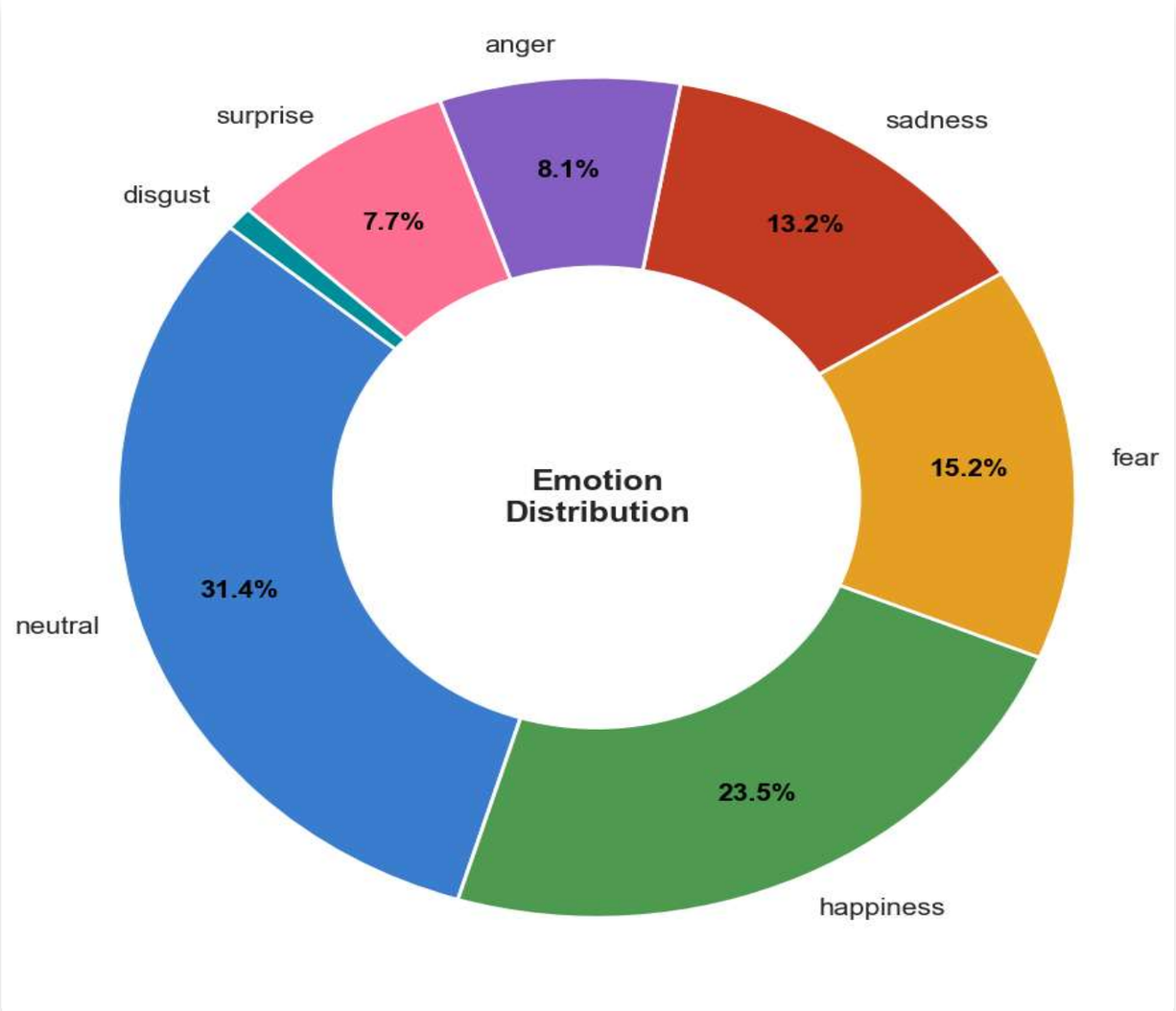
Key Takeaways

- AssemblyAI produced **fewer transcription errors**
- AssemblyAI chosen for pipeline due to **higher accuracy**

Data Characteristics

Key Insights:

- 665 sentences, clean, no missing or duplicate data.
- Emotion imbalance.
- Intensity skewed to mild/moderate.
- Sentences are short, ~8 words.



Data Comparison (Data 1(~73k) vs Data 2(~26k)

Data 1 – 73K samples (Friends dialogue)

- Long, spoken-style sentences
- Mixed or unclear emotions
- Hard for the model to understand

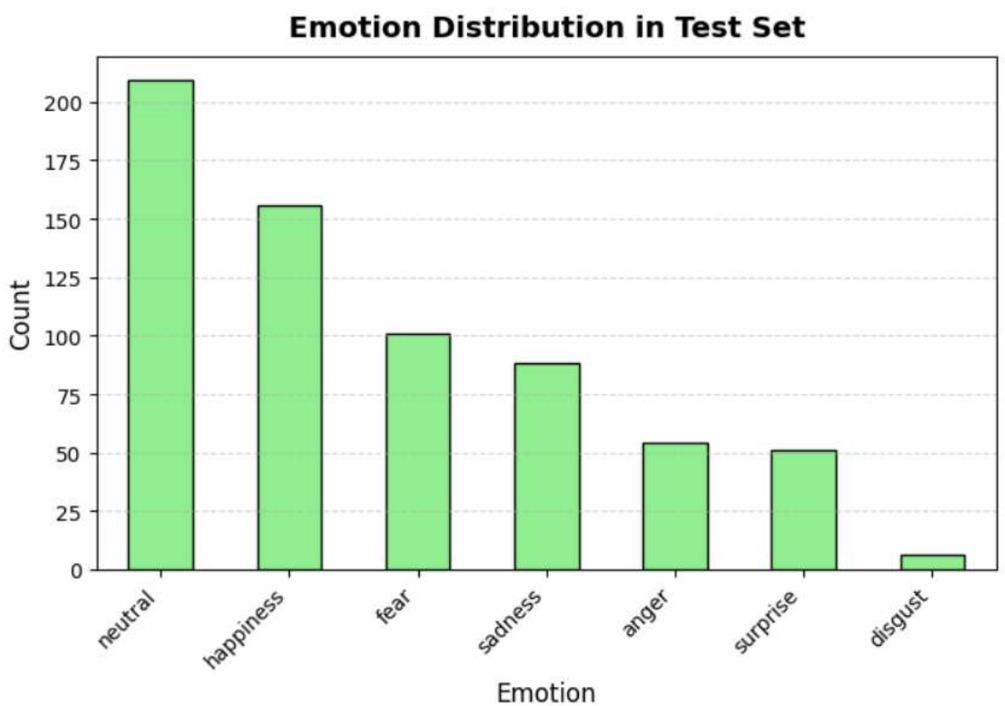
294	: Ahh, thanks but no. You see I-I think IÄ'm ready	fear
295	I can, I can see that.	neutral
296	Umm, but I think if you give me umm, one chance	fear
297	Okay. IÂ'll give you one chance to change my mind. You got one minute.	anger
298	Oh well, itÂ's not on TV yet.	neutral
299	Well, then itÂ's not on the wall yet.	neutral
300	Okay, fine, I will bring you a tape, huh?	neutral
301	So umm, now do you have any of Matt LauerÂ's clothes here? Maybe? Just ones that havenÂ't	neutral

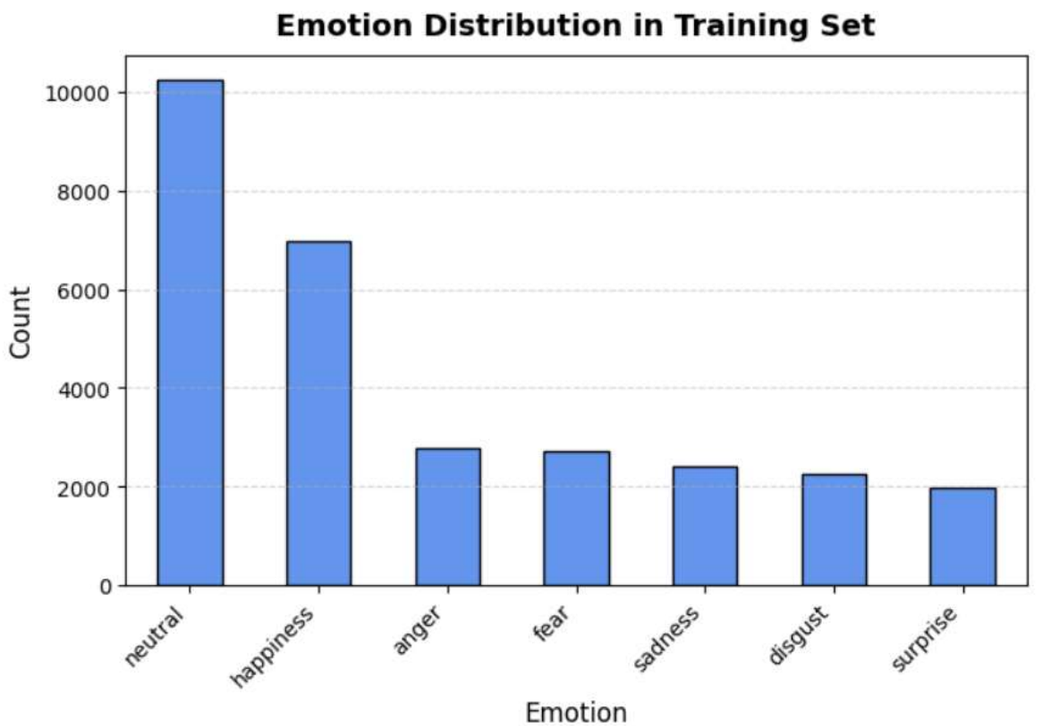
Data 2 – 26k samples

- Short, clear sentences
- Each line shows one strong emotion
- Uses emotional words directly (sad, angry, happy)

35	I can't focus on the game today.	sadness
36	We're already out.	surprise
37	The cool WU likes to wear tank tops, the good one.	happiness
38	If this is cooked beautifully, then you're back to win.	happiness
39	Ah!	happiness
40	This is the granddaughter of Serbian.	neutral
41	You're the reason we lost.	anger
42	It's too dark tonight.	fear

Training set (~26k) and Test set Comparison





Traditional Models

Data Setup

- Cleaned the data
- Added sentiment and POS features
- Split: 80% train / 20% validation, client test-set (≈600 rows)
- Data size: ~26K samples after cleaning

- Logistic Regression
- Naive Bayes
- Support Vector Machine (SVM)

Model	Features	Test F1 (70K)	Test F1 (24K)
Logistic Regression	TF-IDF	0.24	0.36
Naive Bayes	TF-IDF	0.10	0.20
SVM (LinearSVC)	TF-IDF	0.26	0.38
Logistic Regression	TF-IDF + Sentiment	0.27	0.38
Naive Bayes	TF-IDF + POS	0.13	0.29
SVM (LinearSVC)	TF-IDF + Sentiment	0.27	0.39
SVM (LinearSVC)	**TF-IDF + Bigrams + Sent + POS**	**0.29**	**0.42**

Deep learning models

Model	Features	Accuracy	Macro F1 score	Weighted F1 score
RNN	TF-IDF + Sentiment + POS	0.46	0.39	0.45
LSTM	TF-IDF + Sentiment + POS	0.52	0.50	0.52

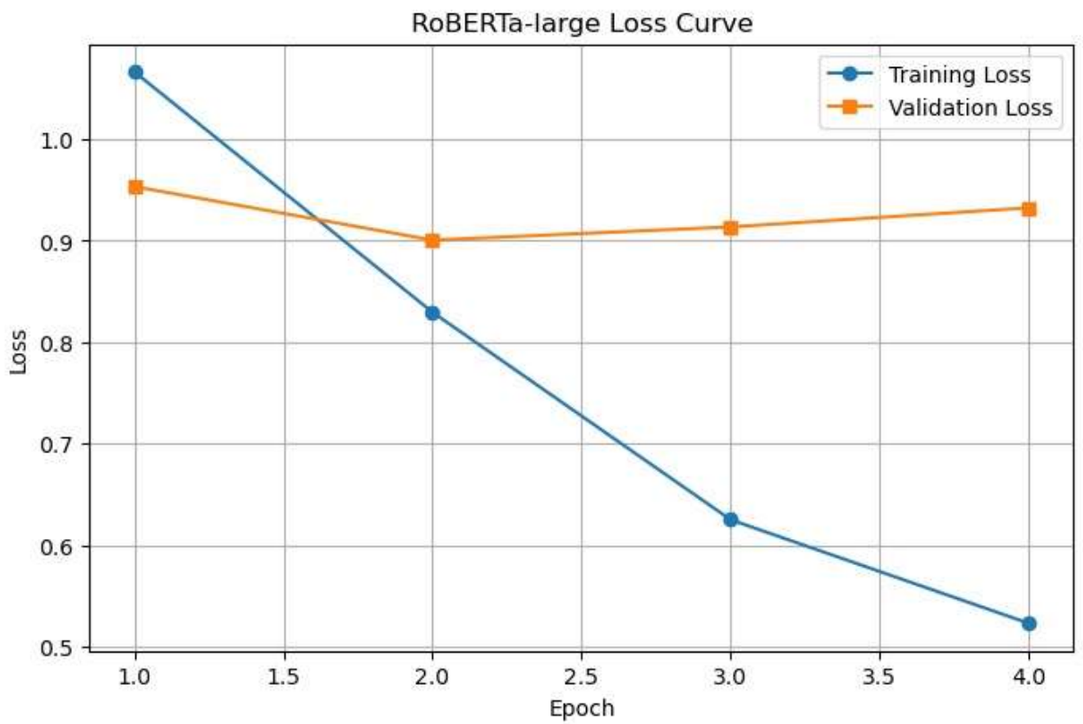
Transformers

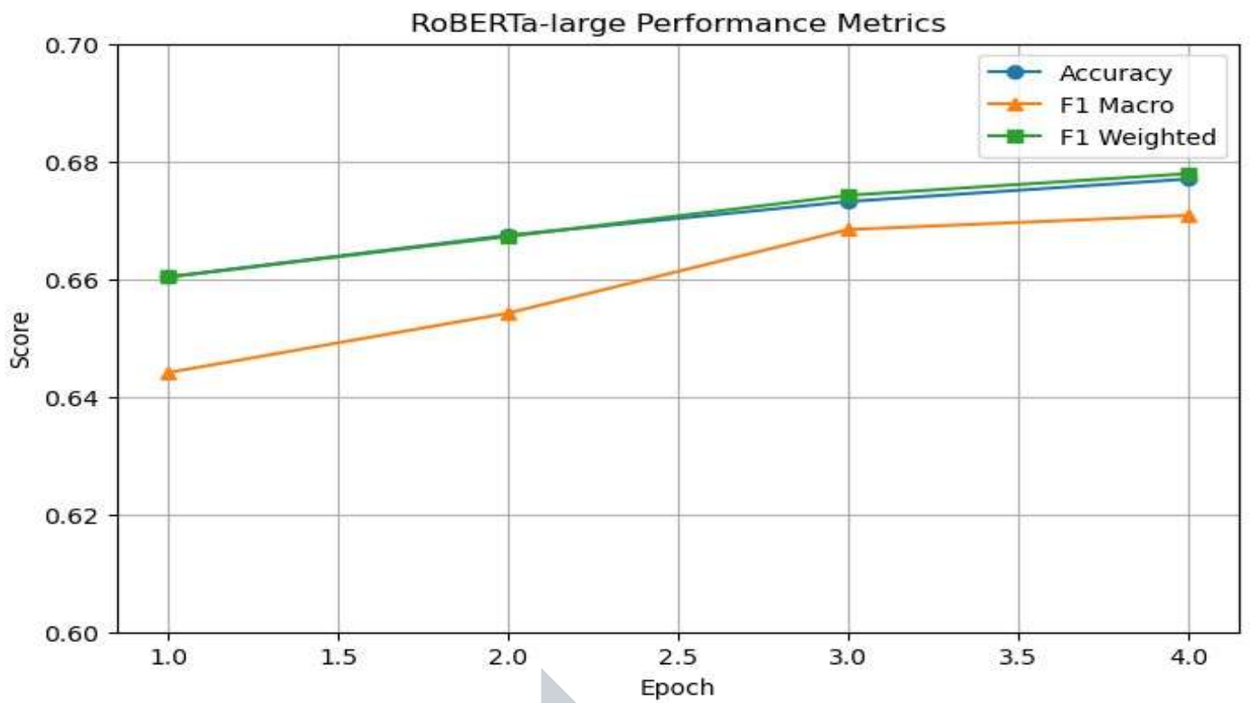
- Best model: **Weighted RoBERTa-Large (**Acc 0.69, F1 0.63)
- RoBERTa-Large close behind (Acc 0.67)
- DeBERTa-v3-Large good but below RoBERTa
- RoBERTa > BERT — better pretraining boosts results
- Class weighting improved emotion balance

Model	Accuracy	Test Macro F1	Test Weighted F1
BERT-base	0.59	0.46	0.59
RoBERTa-base	0.63	0.50	0.63
RoBERTa-Large	0.67	0.60	0.67
Weighted RoBERTa-Large	**0.69**	**0.63**	**0.69**
DeBERTa-v3-Large	0.65	0.55	0.65

Transformers

Weighted RoBERTa-Large Training Progress





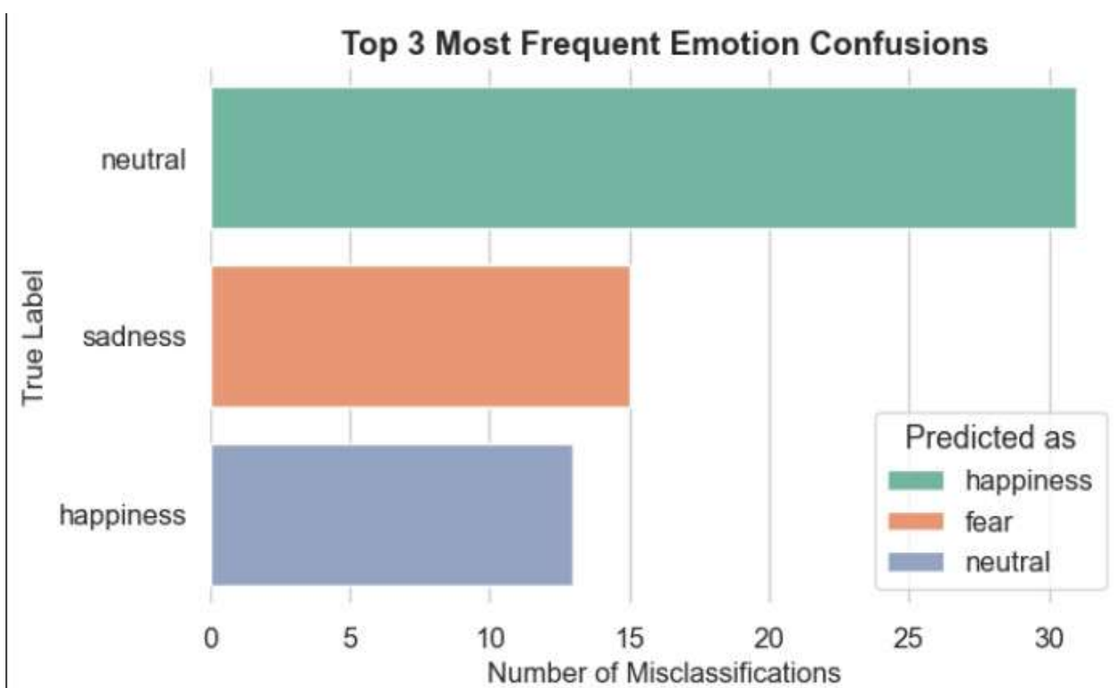
- Model trained for 4 epochs with class-weighted loss
- Training loss steadily decreases — stable learning
- Validation loss stays flat — no overfitting
- Accuracy and F1 scores improve each epoch
- Final performance: Accuracy ~ 0.693, Macro F1 ~ 0.692, Weighted ~ F1 0.692

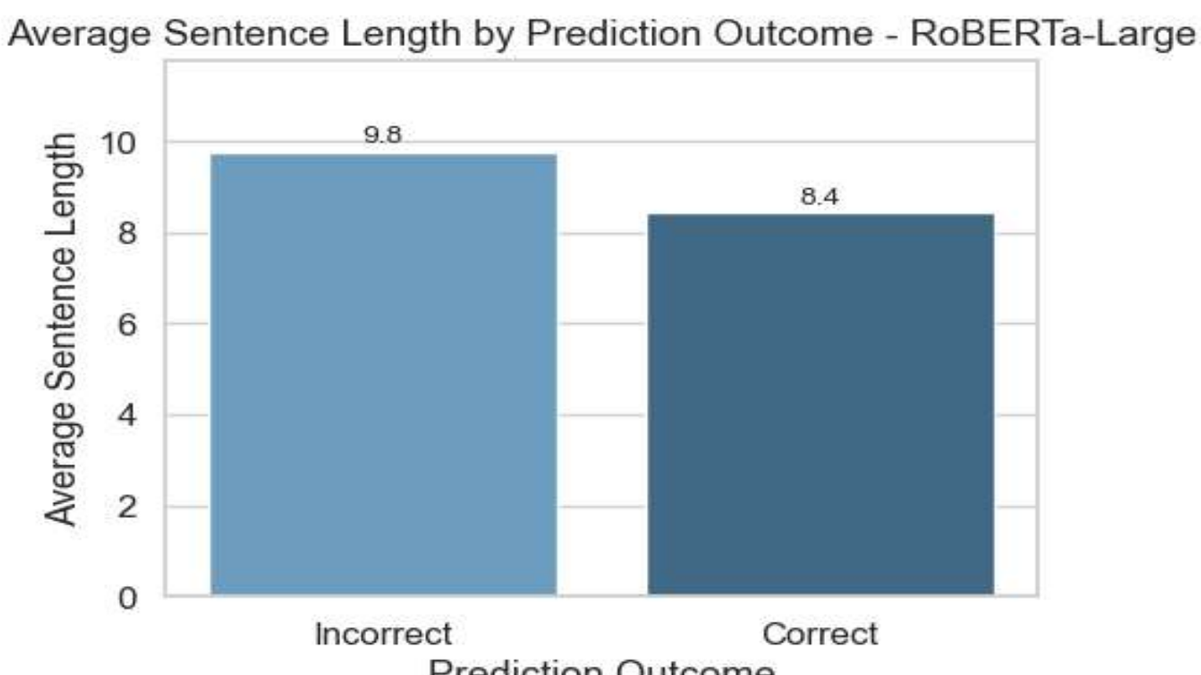
Error Analysis

Common mix-ups between similar emotions

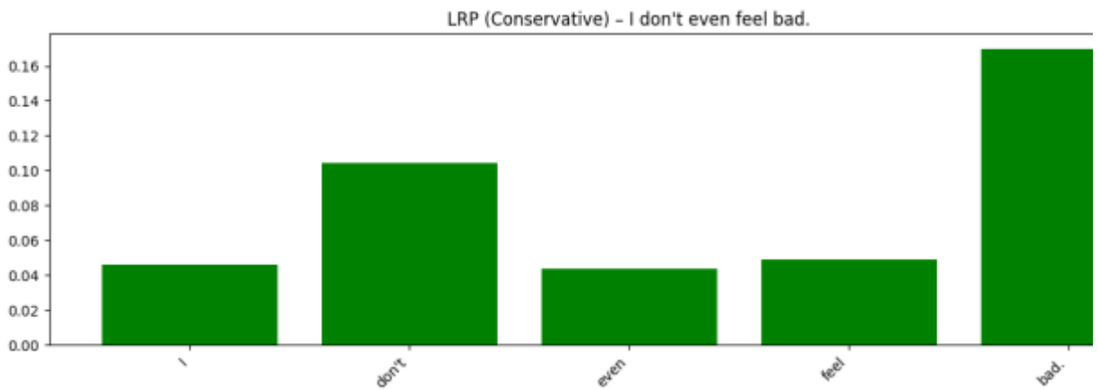
Wrong predictions = longer sentences (~10 words)

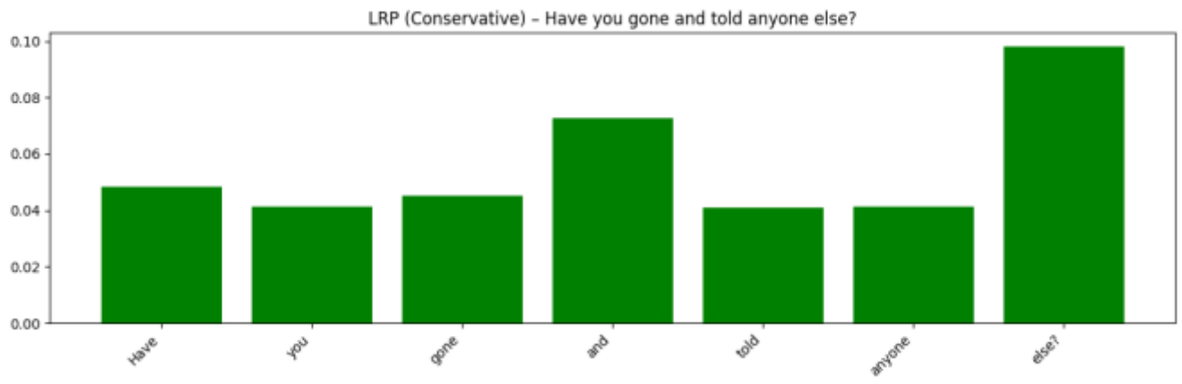
Model strong overall, but needs better context & more rare emotion data





How the model makes its decisions





- The captures strong emotional cues effectively
- Struggles with more neutral and mixed sentences.
- Low confidence

Method / Iteration	Goal (what it fixes)	Macro-F1 (Dev)	Macro-F1 (Test)	Macro-F1 (Full)	Key Notes
Zero-shot Baseline	Simple reference; set a floor	0.540	0.493	0.532	Fast, no context; tends to over-predict *neutral*.
Method A: Step-Back + Escalation	Cut false neutrals via 2-pass + no-neutral backup	0.521	**0.620**	0.542	Conservative JSON parsing; escalate only with affect cues.
Method B: Combo + Calibration	Add retrieval + priors + temp scaling	**0.842**	0.543	**0.772**	Strong overall; dev-tuned, more sensitive than A.
Method C: Hybrid (B→A)	Blend B's strength with A's caution	**0.809**	**0.628**	**0.773**	Trust B when confident; defer to A on neutral/low-conf; H↔S resolver.

Prompt Engineering

- **Data split:** 80% DEV / 20% TEST to tune prompts/hyperparams on DEV.
- **No leakage:** separate cache files per split, and all rules/gates/temperature calibrated on dev only.
- **Overfitting risk:** minimized via simple, interpretable prompts/rules.

Implementation limitations

- Emotional Detection is Subjective
  - Emotions vary by culture, acting style, sarcasm, and personal perception
- Quality of input
  - The quality of the video and amount of background noise, overlapping voices and accents lower the transcription quality
  - The less common classes of emotions are harder for the model to classify leading to lower accuracy
- Privacy and Legal issues

Next Steps

Automatically running the pipeline through youtube URLs

Model improvements as more domain specific data immerge.

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References and questions

- **Ekman, P., & Friesen, W. V. (1971).**

Universals and cultural differences in the judgments of facial expressions of emotion. *Journal of Personality and Social Psychology, 17*(2), 124-129.

- **Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2023).**

Robust Speech Recognition via Large-Scale Weak Supervision. *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, PMLR 202, 28492–28518. Table 13 (“BLEU scores on CoVoST2”), p. 24.